DSP. IV   Task 4.1.

1) $y[n] = 0{,}5\,(x[n] - x[n-1]) + 0{,}8\,y[n-1]$ ← IIR (forever loop) 1st order (feedback of 1 in y and x: $y[n-1]$ ..)

for $n<0$: $y[n]=0$; impulse response $\Rightarrow \delta[n] := x[n] \Rightarrow \begin{cases} 1, & n=0 \\ 0, & \text{else} \end{cases}$

- $n=0$: $h[0] = 0{,}5\,(1-0) + 0{,}8 \cdot 0 = 0{,}5$
- $n=1$: $h[1] = 0{,}5\,(0-1) + 0{,}8 \cdot 0{,}5 = -0{,}1$
- $n=2$: $h[2] = 0{,}5\,(0-0) + 0{,}8 \cdot (-0{,}1) = -0{,}08$
- $n=3$: $h[3] = 0{,}5\,(0-0) + 0{,}8 \cdot (-0{,}08) = -0{,}064$
- $n=4$: $h[4] = \underbrace{\overset{\to b}{0{,}5}\,(0-0)}_{x[n] \text{ always } = 0} + \overset{\to a}{0{,}8} \cdot \underbrace{(-0{,}064)}_{} = -0{,}0512$
- $n \in [5;\infty)$: $h[n] = \underset{\hookrightarrow = 0}{\beta} + 0{,}8 \cdot \alpha \to 0$, system decays by speed of $0{,}8^n$, $n \in [1,\infty)$; it's Infinite Impulse Response (never ends), stable → only decays

2) System's Transfer Function $H(z)$:

shift property: $Z\{x[n-1]\} = z^{-1} X(z)$; $Z\{x[n]\} = X(z) = \sum_{n=0}^{\infty} x[n]\,z^{-n}$

$Z\{y[n-1]\} = z^{-1} Y(z)$

Substitutions: $Y(z) = 0{,}5\,X(z) - 0{,}5\,z^{-1} X(z) + 0{,}8\,z^{-1} Y(z)$

To find TF we need to bring it to the form of $H(z) = Y(z)/X(z)$

$$Y(z)\left(1 - 0{,}8\,z^{-1}\right) = 0{,}5\,X(z)\left(1 - z^{-1}\right)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{0{,}5\,(1 - z^{-1})}{1 - 0{,}8\,z^{-1}} \;\Big|\cdot \frac{z}{z} = \frac{0{,}5\,(z-1)}{z - 0{,}8}$$

3) • Zeros (Roots of numerator): $0{,}5(z-1) \overset{!}{=} 0 \Rightarrow z = 1$

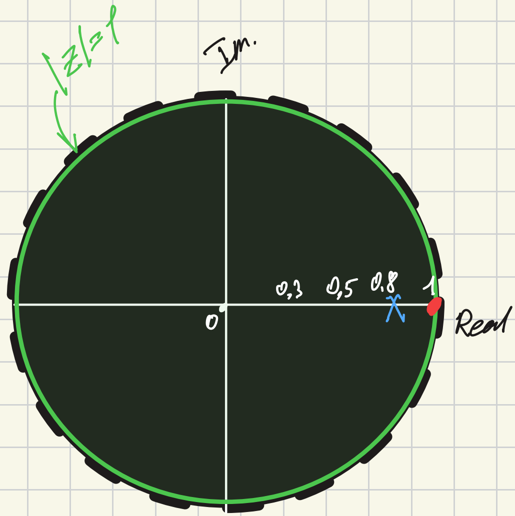
• Poles (Roots of denominator): $z - 0{,}8 \overset{!}{=} 0 \Rightarrow z = 0{,}8 \Rightarrow |z| < 1$ sys. stable

It acts as highpass filter

↳ $z = e^{j\omega}$

- at DC (0Hz): $\omega = 0$: $z = e^0 = 1 \Rightarrow H(1) = \frac{0}{0{,}2} = 0$ → magnitude is blocked
- at Nyquist (limit): $\omega = \pi$: $z = e^{j\pi} = -1 \Rightarrow H(-1) = \frac{-1}{-1{,}8} \approx 0{,}55$ → magnitude is passed

Made with Goodnotes



x Pole: $z = 0{,}8 + j0$

⊙ Zero: $z = 1 + j0$

both on Real axis only.

- Type change (High-pass → Low-pass): move zero from 1 to -1 ⇒

$$H(z) = \frac{0{,}5(z-1)}{z-0{,}8} \Rightarrow \frac{0{,}5(z+1)}{z-0{,}8}$$, then $z=-1$ for new zero*:

$$\frac{Y(z)}{X(z)} = \frac{0{,}5 + 0{,}5z^{-1}}{1-0{,}8z^{-1}} \Rightarrow Y(z) - Y(z)0{,}8z^{-1} = X(z)0{,}5 + X(z)0{,}5z^{-1}$$

$$y[n] - 0{,}8y[n-1] = 0{,}5(x[n] + x[n-1])$$

$$y[n] = 0{,}5(x[n] + x[n-1]) + 0{,}8y[n-1]$$

- Filter class: IIR → FIR (removing recursive loop):

$$y[n] = 0{,}5(x[n] - x[n-1])$$

- Stable → Unstable: we can adjust the feedback (a) param:

$$y[n] = 0{,}5(x[n] - x[n-1]) + 1{,}1\,y[n-1]$$

So, pole located now at $z = 1{,}1 > |z| = 1$.